The finite starting cases from balanced pure cuts

Human Compression project

We acquire the proved interval of pure perfect cuts, its actual contraction and subdivision operations, the actual-wing bound and the exact pure-leaf criterion. A marking is fractured when every maximal omission is intransitive. All equivalences of rooted permutation markings below are point conjugacies taking root to root and preserving the order of the marks. In particular degree-six wings are not identified under the exceptional outer automorphism of S_6.

Call a rooted marking of degree $2p$ and rank p a *wing* if it is transitive, fractured and C, its root is moved by its first mark and fixed by all later marks, and adjoining a pure transposition from the root to a new point gives a C-marking. Write w_p for the number of its rooted classes. The orientation of a wing starts at its root; the left wing in a joined marking is therefore reversed.

Theorem 1. *For $\kappa \geq 1$, the rank-$\kappa+3$ C-markings of $\mathrm{Sym}(2\kappa+3)$, up to marked-group isomorphism and duality, are in bijection with unordered pairs of wings of ranks $p, q \geq 1$ satisfying $p+q=\kappa+1$. In particular their number is*

$$c_\kappa = \sum_{\substack{p<q \\ p+q=\kappa+1}} w_p w_q + \begin{cases} w_h(w_h+1)/2, & \kappa+1=2h, \\ 0, & \kappa+1 \text{ odd.} \end{cases} \tag{1}$$

Proof. Put $n=2r-3$. For $n \geq 7$ the acquired high-rank theorem makes every natural symmetric marking fractured, and the acquired pure-cut theorem gives an internal pure perfect cut. All pure perfect cuts form one interval. At a cut i with prefix support A_i put $d(i)=2i-|A_i|$. Its suffix surplus is $1-d(i)$. Consecutive pure cuts add one point and one mark to the prefix, so $d(i+1)=d(i)+1$.

At the first pure cut we have $d \leq 0$. Otherwise its prefix is symmetric by the actual-wing bound, and its old pure-leaf extension satisfies the full kernel tests. The acquired backward propagation lemma makes the preceding mark pure. Deleting the pendant endpoint leaves the preceding prefix transitive, so this is a pure perfect cut of the whole marking, a contradiction. A positive surplus requires prefix rank at least two, so the preceding label exists. Reversal shows that the suffix surplus at the last pure cut is nonpositive; there $d \geq 1$. There is exactly one consecutive pair with surpluses $0, 1$.

Its adjacent geometry consists of disjoint supports $A, \{c\}, B$, preceding and following groups L, T, and marks $(a\ c), (c\ b)$. If L, T have ranks p, q, the two surplus equations give $|A|=2p$, $|B|=2q$, and $p+q=r-2$. Both ranks are positive, so both cuts are internal. Orienting L from its last mark gives its root a; orient T from its first mark with root b. Distant commutation fixes each root under every later wing mark. Each wing is C by the original IP, and its pure-leaf extension is an old interval. Each is fractured: an omission transitive on its wing would remain transitive on the entire graph through the two bridges and the other wing, contradicting fracture of the parent. Thus these are wings in the stated sense.

Conversely take any two wings on disjoint supports A, B rooted at a, b. Join them first by $q=(a\ b)$. Their pure-leaf extensions ensure that intervals ending or beginning at q are C. For intervals crossing q, the actual one-bridge intersection rectangle from the acquired contraction proof and induction on interval length give IP. For clarity that rectangle is

$$\langle P_0, Q_1, q\rangle \cap \langle P_1, Q_0, q\rangle = \langle P_0, Q_0, q\rangle \quad (P_0 \leq P_1 \leq \mathrm{Sym}(A),\ Q_0 \leq Q_1 \leq \mathrm{Sym}(B)).$$

The full transitive wings generate $\mathrm{Sym}(A \cup B)$ with q. The mark q is a pure perfect cut; its acquired subdivision gives $(a\ c), (c\ b)$ at a new point and preserves IP. The resulting marking has degree $2p + 2q + 1$, rank $p + q + 2$, and its two indicated pure cuts have surpluses $0, 1$. They are therefore precisely its unique pair just described, even if the pure interval is longer. These two constructions are inverse on ordered rooted wing classes.

An ordered point conjugacy must preserve the unique balanced pair; it restricts to rooted conjugacies of its two wings. Conversely these rooted conjugacies extend by sending the middle point to the middle point. Reversal exchanges the two wings. For degree at least seven, marked-group isomorphisms of natural symmetric groups are precisely point conjugacies, by the acquired automorphism argument. In degree five the acquired simplex class is obtained from the unique degree-two, rank-one wing twice. This handles $\kappa = 1$ separately and proves (1) in every case. □

Lemma 2. *The following finite search lists each wing of a fixed even degree n exactly once up to rooted point conjugacy.*

Proof. Set $p = n/2$ and normalize the root to 1. List every nonidentity involution of $\mathrm{Sym}(n)$ by conjugating the products of j disjoint transpositions, for $1 \le j \le n/2$. The first mark moves 1; all subsequent marks fix 1. At each prefix try involutions outside the group already generated, commuting with every earlier mark except the last, and not commuting with that last mark.

This last requirement loses no transitive anchored string marking. The component of its noncommutation diagram containing the first mark is transitive: every other component fixes the root and commutes with that component, and therefore fixes its root orbit pointwise. The full orbit is consequently that component's orbit. Every other component would then consist of identity permutations, which is impossible. Thus the noncommutation diagram is the whole path.

Use one orbit representative under the point stabilizer of 1 at the first step, and thereafter under its subgroup centralizing the entire chosen prefix. This gives exactly the rooted conjugacy classes of prefixes, recursively. Every condition imposed is invariant under these conjugations. Test IP of each new interval by the standard end criterion, using the previously verified intervals. Thus all prefixes retained are C. A separate terminal oracle checks all pairs of all subsets on each retained transitive marking.

A fractured final marking must have every prefix mark essential to the prefix orbit partition. If deleting that mark left the partition unchanged, adding the remaining marks would merge the same blocks in either case and leave the final omission transitive. Reject exactly such inessential prefixes. If a prefix generates H and has length l, also reject when $[\mathrm{Sym}(n) : H] < 2^{p-l}$: independence of each remaining mark would multiply the order by at least two.

For every enlarged prefix P_j, compute $X_{j-1} = 1^{P_{j-1}}$ and impose the actual subgroup inclusion

$$P_j \cap \mathrm{Sym}(X_{j-1}) \le P_{j-1}. \tag{2}$$

These are exactly the acquired full leaf-kernel tests, not tests on an unproved quotient action. Finally retain only transitive terminal markings. Prefix IP, orbit-essentiality and (2) then give respectively C, fracture and pure-leaf validity. Every possible wing survives every necessary test, while rooted orbit representatives eliminate precisely the declared equivalence. For efficiency the candidate list at a prefix retains just the root-fixing involutions commuting with all marks except the last. At length one it is the full root-fixing list; the child list is obtained by imposing commutation with the old last mark. This inductive invariant reproduces the literal distant-commutation test. This proves completeness and uniqueness. □

Theorem 3. *The first six wing counts are*

$$(w_1, \ldots, w_6) = (1, 1, 6, 3, 11, 19).$$

Consequently the first six stable constants in the acquired counting theorem are $(1, 1, 7, 9, 35, 48)$.

Proof. The complete executions of Lemma 2 give the following table. The prefix counts include the empty prefix. An intransitive terminal prefix is not a wing and contributes only to that column.

n	involutions	root-fixing	prefixes by length	wings
2	1	0	$1, 1$	1
4	9	3	$1, 2, 2$	1
6	75	25	$1, 3, 12, 11$	6
8	763	231	$1, 4, 53, 95, 29$	3
10	9495	2619	$1, 5, 187, 1054, 415, 156$	11
12	140151	35695	$1, 6, 552, 10218, 7912, 3149, 617$	19

Every run terminated with its unique completion marker and empty error output. The largest completed run took about 91 seconds. Two earlier 180-second caps supply no part of the count; their outputs are retained. The final candidate-list cache preserves the literal distant-commutation predicate and reproduced the entire degree-ten output byte for byte. Moving the terminal transitivity test before the separate all-subset IP oracle only discards intransitive groups outside the target; every retained wing still receives that full oracle. A further proposed necessary-prefix test made no additional rejections on its control and is not used in the chosen final engine.

The full literal representatives are retained. Independently, a labelled degree-six search tries every root-moving first involution and every root-fixing second and third involution, imposing distant commutation, then checks actual transitivity, all maximal omissions, every subset intersection and each full leaf kernel. It uses neither rooted orbit pruning, the prefix orbit test, the order bound nor the incremental IP test. Its surviving rooted classes agree exactly with the main six. A separate permutation-array implementation checks every subgroup closure and leaf kernel in degrees two, four and six.

A rooted connected marking has a canonical literal array obtained by breadth-first traversal from its root, trying the labels in their fixed order. Two such arrays coincide exactly when a rooted point conjugacy exists. This independently verifies distinctness of every wing list. For each unordered pair in Theorem 1, construct its actual joined tuple. Minimizing the analogous code over every point and the two orientations verifies that all resulting boundary tuples are distinct modulo point conjugacy and reversal. There are respectively

$$1, \quad 1, \quad 6+1=7, \quad 3+6=9, \quad 11+3+21=35, \quad 19+11+18=48.$$

Their generation and IP follow from the proved gluing construction; separate finite controls check these actual tuples as well. Completeness is Theorem 1 together with Lemma 2, not agreement with the historical sequence. The acquired stability theorem transfers these boundary counts to every degree in the stated stable ranges. □

Attribution and scope. The program is an explicit derivative of the previously reviewed degree-ten anchored search, with full rather than even leaf kernels, all first involutions, and all transitive terminal groups. The structural balanced-pair argument and its use to compute the finite bases are new to this recovery, without an assertion of historical priority. The original stable counts remain attributed to Cameron, Fernandes and Leemans and their preceding collaborators. No external atlas completeness claim is used. All actual source inputs, completed runs, capped attempts and changes of engine are pinned separately. This component evaluates the original six constants; it does not begin a further sequence search, new conjecture programme, Lean development or compression measurement.